CS & IT ENGINEERING

Discrete Maths
Set Theory

Lecture No.2

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TOPICS TO BE COVERED

01 Basics operations in sets

02 set theory laws

03 Different operations thm

04 Infinite union

05 Union on intervals

1) union $(\cup)$

2) Intersection $(\cap)$

3) Set difference $(-)$

**4) Symmetric difference $(\oplus / \Delta)$

$A =$

$A = \{1, 2, 3\} \quad B = \{2, 3\}.$

$A \cup B = \{1, 2, 3\}$

$B \cup A = \{1, 2, 3\}$

$A \cap B = \{2, 3\}$

$B \cap A = \{2, 3\}$

$A - B = \{1\}$

$$\begin{array}{cc} A & B \\ 1 & \cancel{2} \\ \cancel{2} & \cancel{3} \\ \cancel{3} & \end{array}$$

$A = \{1, 2, 3\}$ $B = \{2, 3, 4\}$

$A - B = \{1\}$

$B - A = \{4\}$

A	B
1	~~2~~
~~2~~	~~3~~
~~3~~	4

B	A
~~2~~	1
~~3~~	~~2~~
4	~~3~~

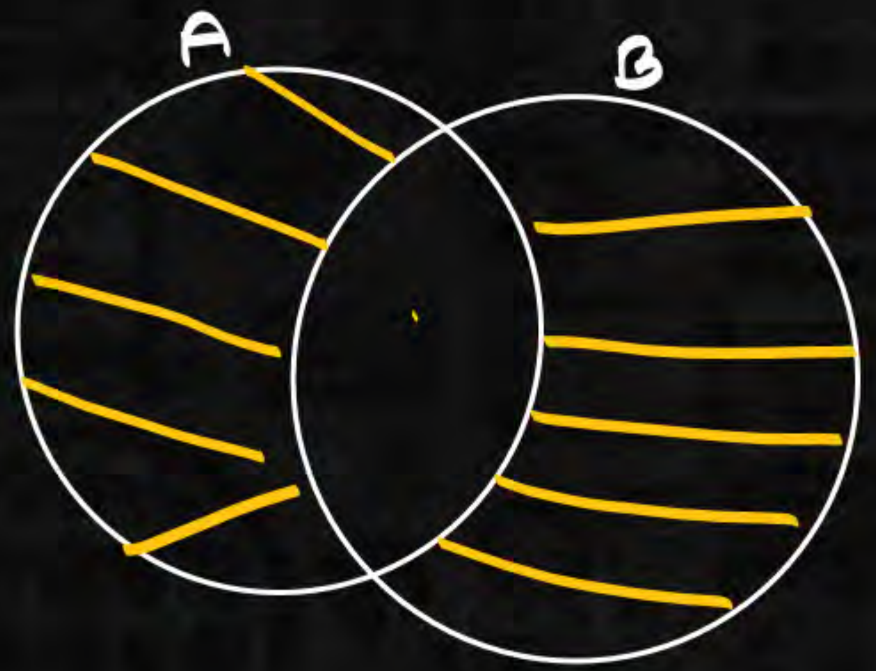
$A - B \neq B - A$

$A - B =$

$$A \triangle B = A \oplus B = (A-B) \cup (B-A) = (A \cup B) - (A \cap B)$$

elements which are present in A, B but not in both.

$$A = \{1, 2, 3\} \quad B = \{2, 3, 4\}$$



$$A \triangle B = \{1, 4\}$$

$$B \triangle A = \{4, 1\}$$

$$A \triangle B = B \triangle A$$

$$\begin{array}{ccc} A & & B \\ 1 & & \cancel{2} \\ \cancel{2} & \cup & \cancel{3} \\ \cancel{3} & & 4 \end{array}$$

$1\,2\,3\,4\,5 \cup 4, 8 = \{1\,2\,3\,4\,5\,8\}$

$A = \{1\ 2\ 3\ 4\ 5\}$

$B = \{1\ 2\ 4\ 8\}$

$C = \{1\ 2\ 3\ 5\ 7\}$

$D = \{2\ 4\ 6\ 8\}$

1) $A \cup (B - C)$

2) $(B - C) - D$

3) $B - (C - D)$

4) $(A \cup B) - (C \cap D)$

$(B - C) - D = \phi$

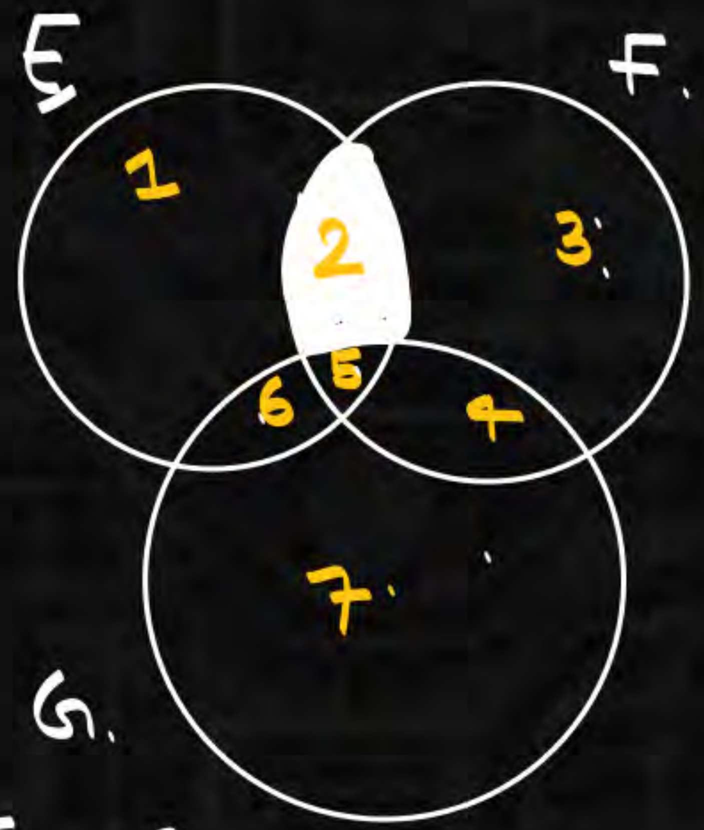
$$\begin{array}{ccc} \cancel{1} & & \cancel{1} \\ \cancel{2} & - & \cancel{2} \\ 4 & & 3 \\ 8 & & 5 \\ & & 7 \\ \hline \end{array}$$

$$\begin{array}{ccc} \cancel{4} & - & 2 \\ \cancel{8} & & \cancel{4} \\ & & 6 \\ & & \cancel{8} \end{array}$$

$x = (E \cap F) - (F \cap G) = \underline{\text{part } 2}$

$y = \left(E - (E \cap G)\right) - (E - F)$

(GATE)



partwise distribution.

$\cancel{\begin{matrix}2\\5\end{matrix}} - \cancel{\begin{matrix}4\\5\end{matrix}} = \{2\}$

a) $x = y$

b) $x < y$

c) $y < x$

d) nota.

E		F
1		2
2		3
5	$\cap$	4
6		5

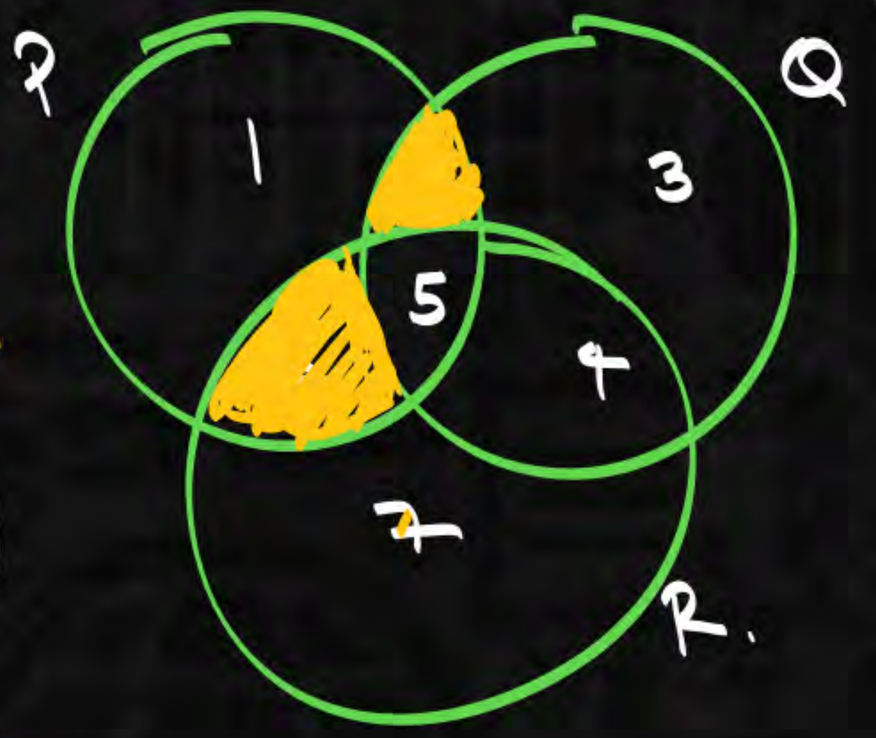
$= 2$
5

F		G
2		4
3	$\cap$	5
4		6
5		7

$= 4\,5$

I

$$\begin{array}{c} 1 \\ 2 \\ 5 \\ 6 \end{array} \cap \begin{array}{c} 2 \\ 3 \\ 4 \\ 5 \end{array} = 25 \quad \triangle \quad \begin{array}{c} 1 \\ 2 \\ 5 \\ 6 \end{array} \cap \begin{array}{c} 4 \\ 5 \\ 6 \\ 7 \end{array} = 56.$$



II. $P \cap (Q \triangle R) = (P \cap Q) \triangle (P \cap R)$

$$\begin{array}{c} 2 \\ 3 \\ \cancel{4} \\ \cancel{5} \end{array} \triangle \begin{array}{c} 6 \\ \cancel{5} \\ \cancel{4} \\ 7. \end{array}$$

$$2\cancel{5} \triangle \cancel{5}6 = \underline{\underline{26}}.$$

$$\begin{array}{c} 1 \\ 2 \\ 5 \\ 6 \end{array} \cap \begin{array}{c} 2 \\ 3 \\ 6 \\ 7. \end{array} = 26$$

I. $P \triangle (\boxed{Q \cap R}) = (P \triangle Q) \cap (P \triangle R)$

$\{2, 3, 4, 5\} \cap \{4, 5, 6, 7\}$

$P \triangle Q \cap R$

$\{1, 2, \cancel{5}, 6\} \triangle \{4, \cancel{5}\} = \underline{1\,2\,4\,6}$

$\{1, \cancel{2}, \cancel{5}, 6\} \triangle \{\cancel{2}, 3, 4, \cancel{5}\}$ $\quad$ $\{1, 2, \cancel{5}, \cancel{6}\} \triangle \{\cancel{2}, \cancel{5}, 4, 7\}$

$1\,6\,3\,4 \cap 1\,2\,4\,7$

$1\,4$

$S_1.$ $A-(B\cup C)=(A-B)\cap(A-C)$

$S_2.$ $A\triangle(B\triangle C)=(A\triangle B)\triangle C.$

(Valid)

A, B are sets. valid?

1. $P(A \cup B) = P(A) \cup P(B)$

2. $P(A \cap B) = P(A) \cap P(B)$

valid?

3. if $A \cap B = A \cap C \longrightarrow B = C$

4. if $A \cup B = A \cup C \longrightarrow B = C$.

5. if $A \Delta B = A \Delta C \longrightarrow B = C$.

$$P(A \cup B) = P(A) \cup P(B) \quad (\text{Invalid})$$

$$A = \{1\} \qquad B = \{2\} \longrightarrow A \cup B = \{1, 2\}$$

$$P(A) = \{\phi, A\} \qquad P(B) = \{\phi, B\} \qquad \underline{P(A \cup B)} = \{\phi, \{1\}, \{2\}, \{1, 2\}\}$$

$$P(A) \cup P(B) = \{\phi, A, B\}$$

$A = \{1\}$ $\quad B = \{1, 2\}$ $\longrightarrow$ $A \cap B = \{1\}$

$P(A) = \{\phi, A\}$ $\quad P(B) = \{\phi, \underset{A}{\{1\}}, \{2\}, \{1 2\}\}$ $\quad P(A \cap B) = \{\phi, \{1\}\}$

$P(A) \cap P(B) = \{\phi, A\}$ $\quad = \{\phi, A\}$

Thm.

$$\boxed{P(A \cap B) = P(A) \cap P(B)}$$

if $A \cap B = A \cap C \longrightarrow B = C$.

$$\underbrace{A \cap B}_{\{2\}} = \underbrace{A \cap C}_{\{2\}} \longrightarrow \underline{B = C}.\ \text{(Invalid)}$$

$$\{2\} \cap \{2, 10\} = \{2\} \cap \{2, 11\}$$

(Invalid)

4. if $A \cup B = A \cup C \longrightarrow B = C$.

5. if $A \triangle B = A \triangle C \longrightarrow B = C$.

(valid)

$$\overbrace{A \cup B}^{\{2\ 3\}} = \overbrace{A \cup C}^{\{2\ 3\}} \longrightarrow B = C.$$

$$\underbrace{\{2\} \cup \{3\}}_{\{2\ 3\}} \qquad \underbrace{\{2\} \cup \{2\ 3\}}_{\{2\ 3\}}$$

5. $$\overbrace{A \triangle B}^{\{2\ 3\}} = \overbrace{A \triangle C}^{\{2\ 3\}} \longrightarrow B = C.$$

$$\{2\} \triangle \{3\} \qquad \{2\} \triangle \{3\}$$

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$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$
$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (A \cap B) = A$$
$$A \cap (A \cup B) = A$$

$$A \cup A = A$$
$$A \cap A = A$$

$$A \cup U = U$$
$$A \cap U = A$$
$$A \cap \phi = \phi$$
$$A \cap U = A$$

$\wedge$	$\cap$	$\cdot$
$\vee$	$\cup$	$+$
T	U	1
F	ϕ	0

$A_1 = \{ \quad \}$ $A_2 = \{ \dots \}$ $A_3 = \{ \dots \}$

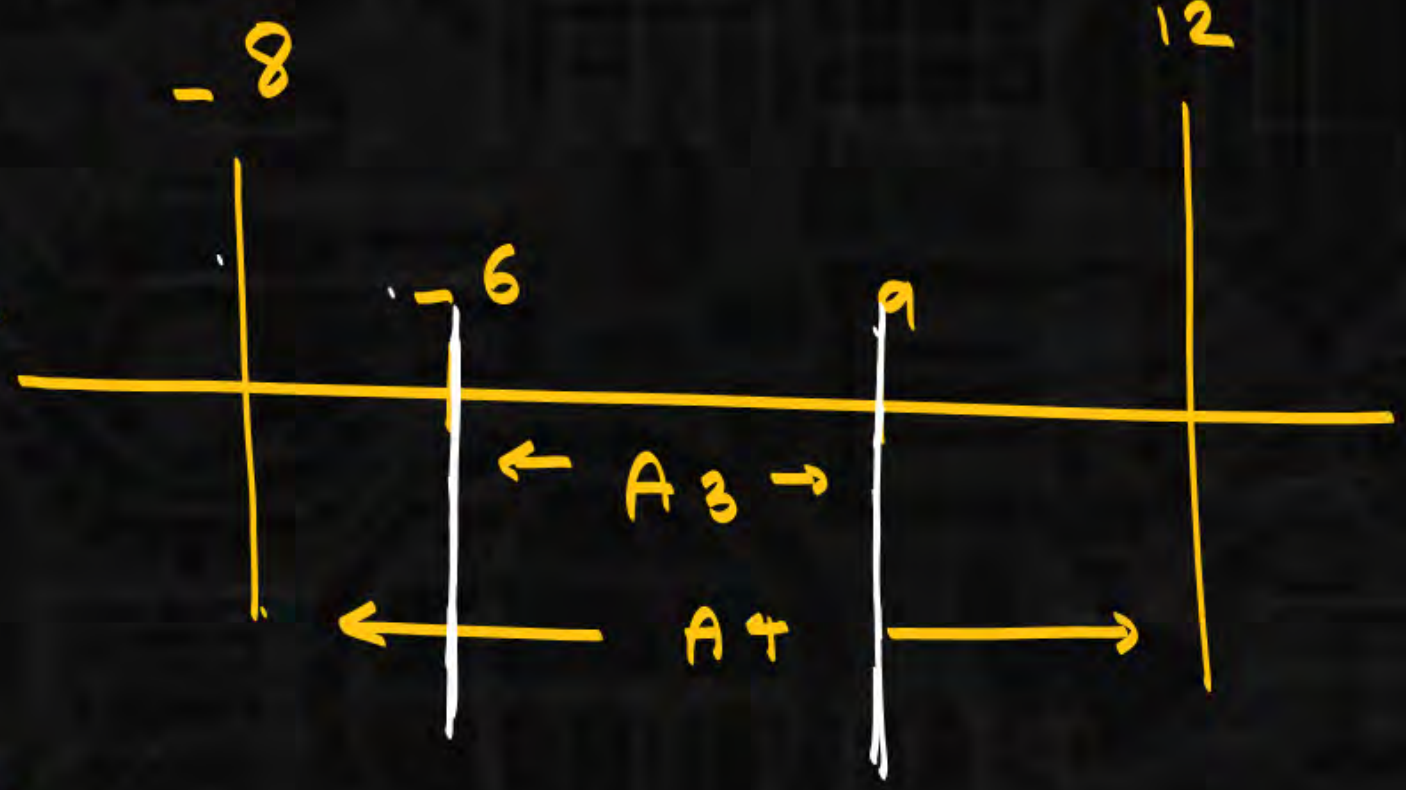
$$A_1 \cup A_2 \cup A_3 = \bigcup_{i=1}^{3} A_i$$

$A_n = [-2n, 3n]$ z.

1. A_3 $[-6, 9]$
2. $A_3 - A_4 = \phi$
3. $A_3 \triangle A_4$

$= [-8, -6) \cup (9, 12]$

4. $\bigcup_{i=1}^{7} A_i = A_7$
5. $\bigcap_{i=1}^{7} A_i = A_1$




THANK YOU GW
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